2025 IEEE 36th International Symposium on Software Reliability Engineering Workshops (ISSREW) | 979-8-3315-5325-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ISSREW67781.2025.00052

Bayesian Network-Based Scenario Generation for Autonomous Systems Testing

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***Abstract*—Scenario-based testing of autonomous systems, such as autonomous vehicles, faces a fundamental challenge: the exponential growth of test scenarios due to interacting variables. Traditional methods either exhaustively test all combinations (infeasible for large systems) or prioritize arbitrary subsets (risking coverage gaps). This PhD work proposes a novel scenario generation framework that systematically addresses these limitations through an integrated approach combining Bayesian Networks, Operational Design Domain (ODD), and enhanced combinatorial testing strategies. First, a Bayesian Network model captures probabilistic dependencies among scenario parameters, including ODD parameters, ensuring generated scenarios are realistic, physically plausible, and reflect real-world distributions. Second, abstracted parameters, which are higher-level constructs that consolidate shared effects from multiple initial variables, are introduced to group scenarios with similar system effects. Third, full combinatorial testing is applied to these abstracted parameters to ensure comprehensive coverage. Finally, for each combination of abstracted parameters, probabilistic inference selects initial variable combinations that achieve the desired abstraction while prioritizing high-criticality scenarios. This approach enables efficient scenario representation, balancing completeness, effectiveness, computational efficiency, and scenario realism without necessitating exhaustive exploration of the entire scenario parameter space.**

***Index Terms*—Scenario-based testing, scenario generation, bayesian networks, combinatorial explosion, verification and validation, autonomous systems.**

I. Research Problem and Motivation

Autonomous systems, as defined by ISO/IEC TR 29119-11:2020, are systems capable of operating without human intervention for extended periods, encompassing physical applications such as autonomous vehicles, unmanned aerial vehicles, and industrial robots [1]. These systems typically comprise three high-level functional components (i.e., sensing, decision-making, and control) where artificial intelligence plays a critical role, particularly in processing and interpreting sensor data.

Given the complexity and safety-critical nature of these systems operating in dynamic environments, scenario-based testing (SBT) emerges as a structured approach for safety assessment by systematically decomposing the operational environment into representative scenarios [2], [3]. The need for such testing methodology becomes evident when considering that autonomous systems must safely navigate diverse and unpredictable real-world conditions, necessitating a systematic scenario-based approach that can effectively challenge the system under various operational conditions within its defined Operational Design Domain (ODD). The ODD specifies the operating conditions under which an autonomous system is designed to operate safely [4]. Defining the ODD is crucial to establishing the operational boundaries within which the system will function and ensure the system's safety within this perimeter [5].

Despite its promising approach, scenario-based testing faces several significant challenges. The main obstacle is the parameter space explosion phenomenon, where the combinatorial possibilities of parameter variations can generate a huge number of scenarios that must be evaluated [6], making exhaustive testing impossible. In autonomous vehicles, environmental conditions, dynamic traffic, and infrastructure variations interact to create millions of possible driving scenarios that must be systematically tested for safety validation. Current research can be categorized into four distinct scenario generation approaches: edge-case-focused, sampling methods, combinatorial testing, and data-driven methods.

Edge-case-focused methods, including search-based approaches with fitness functions (e.g., [11], [12]), reinforcement learning algorithms (e.g., [13]), and boundary-focused techniques (e.g., [14]), employ criticality metrics tailored to specific System Under Test (SUT) components such as trajectory planning, making them unsuitable for the scenario-based black-box testing concept required by the Safety of the Intended Functionality (SOTIF) standard [7]. The SOTIF standard (ISO 21448) addresses failures that are not directly traceable to design faults but arise from the interaction between a system and its operational environment [8]. SOTIF highlights the need for comprehensive scenario-based testing that can uncover unknown unsafe scenarios by achieving high coverage and ensuring broad exploration of the scenario space. Moreover, these edge-case methods serve development and internal validation purposes rather than safety validation [10], as their edge case discovery inherently depends on predefined performance metrics and the tested system's responses, which limits the exploration of unknown scenarios demanded by safety standards.

Sampling methods provide the possibility to respect available parameter distributions and enable parallel test execution since drawn samples are mutually independent, reducing

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DOI 10.1109/ISSREW67781.2025.00052

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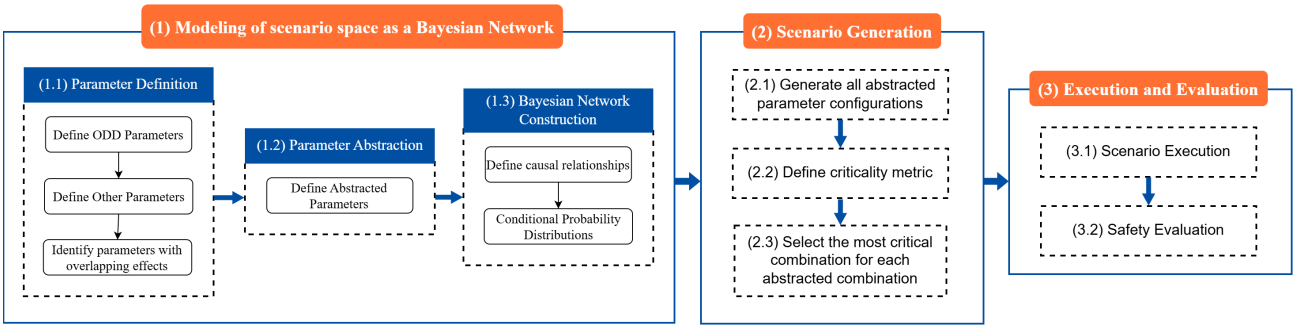


Fig. 1. Workflow of the proposed approach.

overall testing time. However, most sampling methods assume parameter independence and ignore natural correlations (e.g., [15]), distorting scenario realism, while uniform sampling ignores real-world exposure probabilities, leading to inefficient testing of impractical scenario volumes [10]. Traditional combinatorial testing generates excessive redundant cases while failing to capture the complexity and correlations inherent in real-world scenarios. Finally, data-driven methods (e.g., [16]), while addressing some parameter dependencies, remain constrained by the limitations of available datasets and cannot generate scenarios beyond collected data variations.

Hence, the problem remains that existing scenario generation approaches either lack the capability to prioritize critical scenarios in the ODD independently of specific system implementations, fail to preserve realistic parameter correlations and exposure probabilities, lack the capability to systematically explore the complete operational design domain, or are limited by the scope of available real-world data, ultimately hindering the development of comprehensive and efficient testing frameworks that can adequately assess the safety for autonomous systems across their entire ODD.

To address these challenges, this PhD project aims to propose a novel scenario generation framework that integrates Bayesian Networks with ODD modeling and enhanced combinatorial testing strategies. Our approach constructs a Bayesian network to capture probabilistic dependencies among scenario parameters, including ODD parameters, to ensure that generated scenarios are realistic, physically plausible, and reflect real-world distributions. We will then introduce abstracted parameters as higher-level constructs that consolidate shared effects from multiple initial variables. Full combinatorial testing is applied to these abstracted parameters to ensure comprehensive coverage. Probabilistic inference selects initial variable combinations for each abstraction that achieve the desired representation while prioritizing high-criticality scenarios. This integrated methodology transforms the exponential challenge of exhaustive scenario generation into a targeted, statistically informed approach that balances completeness, effectiveness, computational efficiency, and scenario realism without requiring exhaustive exploration of the entire scenario parameter space.

Table I systematically maps how existing approaches fail to address our framework's core capabilities. Edge-case methods lack comprehensive coverage due to SUT-specific fitness functions and convergence toward limited failure modes. Sampling methods ignore realistic parameter correlations through independence assumptions. Combinatorial testing generates scenarios without criticality prioritization or realistic dependencies. Data-driven approaches cannot achieve comprehensive coverage beyond their source datasets. In contrast, our approach uniquely addresses all four capabilities simultaneously.

TABLE I
CAPABILITY GAPS IN EXISTING SCENARIO GENERATION METHODS

Method Category	System-Agnostic	Realism	Coverage	Criticality Prioritization
Edge-case focused	×	×	×	✓
Sampling	✓	×	✓	✓
Combinatorial	✓	×	✓	×
Data-driven	✓	✓	×	✓
Our Framework	✓	✓	✓	✓

II. RESEARCH HYPOTHESIS

Our research hypothesis is that causal dependencies exist between scenario parameters, and a scenario generation framework integrating Bayesian Networks to model these dependencies, abstracted parameters to reduce dimensionality, and probabilistic inference to prioritize critical scenarios will generate more effective and realistic test cases for autonomous systems compared to current approaches, while simultaneously providing deeper insights into failure-triggering conditions and ensuring comprehensive coverage of distinct system effects (i.e., ways scenarios can influence system decisions).

III. PROPOSED APPROACH AND EXPECTED CONTRIBUTIONS

Bayesian reasoning methods offer a mathematically rigorous framework for representation, inference, and learning in complex real-world domains [9], making them ideally suited for test scenario generation. Our approach leverages these capabilities by constructing a Bayesian network that integrates

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the ODD of the system under test with probability distributions over its parameter values. This network explicitly models the conditional dependencies among key ODD parameters, enabling systematic reasoning about the likelihood of various parameter combinations. By employing probabilistic inference, the network efficiently prunes implausible or redundant scenarios, while prioritizing critical ones, thereby mitigating the combinatorial explosion inherent in exhaustive testing.

In addition, our method introduces the novel concept of *abstracted parameters*, which consolidates multiple initial variables with shared effects into higher-level representations, simplifying the overall model while enhancing causal clarity. For instance, rather than treating fog density and fog distance as separate parameters, we recognize that both impact the system's *visibility* and therefore create a single abstracted parameter that captures their combined effect. This abstraction ensures that the testing process captures all meaningful effects in system decisions without incurring unnecessary computational overhead. The overall workflow of this methodology is illustrated in Figure 1.

Achieving the proposed approach requires 3 main contributions:

A. Modeling of scenario space as a Bayesian Network

We will model the autonomous system's scenario space as a Bayesian Network that captures real-world dependencies among ODD parameters. This moves beyond traditional independence assumptions by systematically identifying, quantifying, and encoding probabilistic dependencies among scenario parameters. We will create systematic methods for automatically constructing these networks from the ODD, incorporating both domain knowledge and real-world data to ensure practical applicability. Furthermore, we will propose a method to define the abstracted parameters as higher-level constructs that consolidate the shared effects of multiple initial variables. This abstraction enables a more manageable scenario representation while preserving the essential effects that influence the system behavior.

B. Criticality-based scenario generation

We will develop a scenario generation algorithm that combines comprehensive combinatorial testing with criticality-based selection to optimize scenario effectiveness. Our approach first applies full combinatorial testing to abstracted parameters to ensure complete coverage of distinct effects on the system. For each combination of abstracted parameters, the algorithm then employs probabilistic inference on the Bayesian network to select specific initial parameter combinations that achieve the desired abstraction while prioritizing scenarios with high criticality.

Central to this contribution is the definition of a criticality metric that quantifies the likelihood of scenario-induced failures in specific autonomous functions. This metric should recognize that criticality is not uniform across all scenario types and must be context-dependent, accounting for the fact that parameter combinations that are critical for one scenario may not be critical for others. The algorithm maintains physical plausibility through conditional inference while systematically exploring the space of abstracted parameters, ensuring that no important combinations are overlooked while focusing computational resources on the most safety-critical instantiations of each abstract scenario type.

C. Defining metrics for safety evaluation

Evaluating executed scenarios represents a crucial step in scenario-based testing workflows. However, existing literature reveals that most current safety evaluation metrics suffer from a significant limitation: they cannot fully capture safety's multifaceted nature. Conventional quantitative measures, such as time to collision, often focus solely on worst-case behaviors, leading to severely reduced specificity [10]. To address these limitations, we will develop an advanced evaluation method that enables automated safety assessment of autonomous systems. This method will leverage learned behavioral patterns from historical data to make informed pass/fail decisions. This approach aims to deliver more robust and scalable safety assessments that adapt to the specific characteristics of each evaluated scenario.

IV. Evaluation Methodology

To comprehensively evaluate our proposed method, we will establish evaluation criteria and metrics that enable a holistic comparison with existing approaches in the literature. Our comparative analysis will extend beyond traditional failure detection and coverage measures. Key metrics for this evaluation include, but are not limited to:

- ***Interpretability***. Assessing the transparency and explainability of why specific scenarios are generated, enabling practitioners to understand the underlying reasoning and logic.
- ***Effectiveness***. Maintaining the traditional focus on failure detection capabilities.
- ***Bias***. Evaluating potential systematic biases in scenario generation that could lead to over-representation or under-representation of certain regions in the scenario space.
- ***Informativeness***. Measuring the quality and utility of information provided when system failures occur, particularly regarding the conditions that triggered the failure.
- ***Coverage***. Assessing the extent to which the generated scenarios encompass known critical scenarios identified through domain expertise and historical data.

V. Preliminary Work

To validate the feasibility of our approach, we have conducted a proof-of-concept study on a high-risk driving context: a T-junction scenario. This setup involves an ego vehicle and an adversarial vehicle navigating an intersection with up to three potential collision points. We used the scenario space that was defined in [17], incorporating vehicle paths and environmental parameters such as fog, precipitation, and road friction, resulting in over 136 million potential combinations.

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Using our methodology, we constructed a Bayesian Network to model conditional dependencies and removed implausible parameter combinations, reducing the space to 1.75 million valid cases. We then applied full combinatorial testing on abstracted parameters to generate 648 testable scenarios. To prioritize critical scenarios for each abstracted parameter combination, we used probabilistic inference to select both high- and low-likelihood combinations.

Our early results demonstrate that rare scenarios (low-likelihood) uncover significantly more collisions than two baseline methods: random sampling and SitCov (a state-of-the-art coverage-based method) [17]. Additionally, our method identified low visibility conditions and specific collision points as the dominant failure triggers for the considered ego vehicle model. This demonstrates our method's capability to both generate effective test scenarios and provide interpretable insights into failure conditions.

VI. Current Progress and Timeline

My PhD project has made significant progress across multiple foundational areas essential to the proposed framework. First, we have conducted a comprehensive systematic literature review of scenario-based testing methodologies for autonomous systems, which provides a thorough analysis of existing approaches, their limitations, and research gaps. The findings from this systematic review are currently being prepared for submission and will serve as the theoretical foundation for our contributions. Second, we have developed and implemented initial versions of our Bayesian Network-based scenario generation framework, demonstrating feasibility through T-junction scenario experiments in CARLA simulator. These preliminary experiments have demonstrated the feasibility of our approach and initial results have shown promising improvements in the effectiveness and efficiency of the generated scenarios.

Based on the current progress and the scope of remaining work, we anticipate defending the doctoral thesis in mid-2027. This timeline allows for the completion of all three major contributions outlined in our research plan and comprehensive experimental validation. The remaining research activities include: finalizing the systematic methodology for Bayesian network construction from ODD specifications, completing the development of our criticality-based scenario generation algorithm with robust metrics, implementing and validating our safety evaluation method, and conducting extensive comparative evaluations against existing approaches using the comprehensive metrics we have defined.

Acknowledgment

This research is being conducted at the French Alternative Energies and Atomic Energy Commission (CEA) under the supervision of Nikolas Matragas, Morayo Adedjouma, and Chokri Mraidha. It was partially funded by the European Union's Horizon 2020 research and innovation programme under grant agreement No. 101120990 as part of the project SOPRANO. The content reflects only the authors' views, and the European Commission is not responsible for any use that may be made of the information it contains.

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